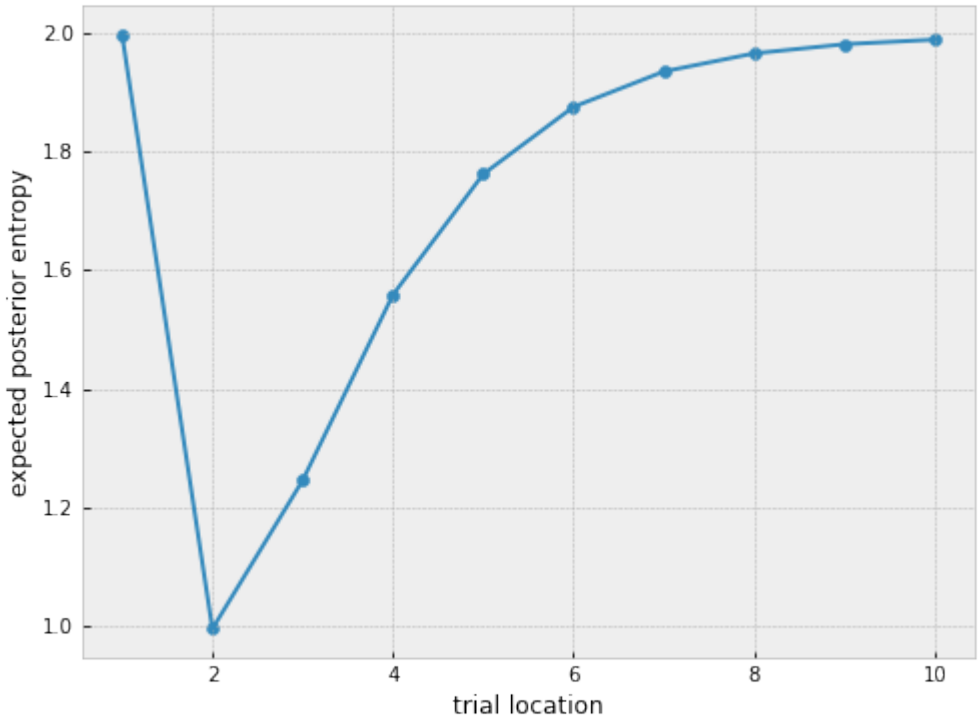


Figure A.14 The expected posterior entropy as a function of the location to conduct the trial.

This plot is shown in Figure A.14, which tells us that the information-theoretically optimal location for our first trial is the 2nd floor, since 2 gives us the lowest expected posterior entropy (and therefore uncertainty).

We see that this is *not* the same as the decision binary search suggests, 5. This is a direct effect of the domain knowledge that we are encoding using the prior distribution for X: since there's a high chance that $X = 2$ (50%), it's actually better to simply try out that number first, in case that we can immediately find our answer if the phone breaks.

Interestingly, dropping the phone from the first floor gives us no reduction in entropy. This is because we know for sure that the phone won't break from this floor, so our knowledge of the world won't change after conducting this trial.

A.6.2 Bayesian optimization for hyperparameter tuning

This exercise, implemented in `CH06/03 - Exercise 2.ipynb` applies BayesOpt to an objective function that simulates the accuracy surface of a support-vector machine model in a hyperparameter tuning task.

1. Recreate the Bayesian optimization loop in `CH04/03 - Exercise 2.ipynb`, including the outer loop that implements repeated experiments.
2. Run the Max-value Entropy Search (MES) policy.
   Since our objective function is two-dimensional, we should set the size of the Sobol sequence used by MES to be 2000.

©Manning Publications Co. To comment go to liveBook
https://livebook.manning.com/#!/book/bayesian-optimization-in-action/discussion

Licensed to ashwin sampath <ashwin.manning@gmail.com>